# GREEEN tallgrass prairie restoration: RLQ analyses (SW Station)

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## Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

### Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types =
    cols(
    .default = "?",
    seeding_time = "f"
```

```r
  )) %>%
  arrange(id_plot_year) %>%
  filter(
    site == "SW Station",
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  filter(treatment != "unseeded_0_0") %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  mutate(seedmass = log(seedmass)) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size %in% c("1", "25")) %>%
  mutate(abundance = 1) %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

### Load data

## Checks

```r
species %>%
  group_by(id_plot_year) %>%
  count()
```

### Seeded species richness

```
## # A tibble: 69 x 2
## # Groups:   id_plot_year [69]
##    id_plot_year     n
##    <chr>        <int>
##  1 SWS_11_2015      6
##  2 SWS_11_2016      7
##  3 SWS_11_2017      8
```

```
##  4 SWS_11_2018      7
##  5 SWS_17_2015      4
##  6 SWS_17_2016      5
##  7 SWS_17_2017      6
##  8 SWS_17_2018      5
##  9 SWS_19_2015      7
## 10 SWS_19_2016      7
## # i 59 more rows
```

```
traits %>% anti_join(species, by = "accepted_name")
```

**Check non-occuring seeded species**

```
## # A tibble: 14 x 4
##    accepted_name                   sla height seedmass
##    <chr>                         <dbl>  <dbl>    <dbl>
##  1 Drymocallis arguta            136.   0.91     -8.86
##  2 Echinacea purpurea            126.   0.37     -6.14
##  3 Eryngium yuccifolium           91.7  1.01     -5.57
##  4 Liatris cylindracea           110.   0.564    -5.73
##  5 Penstemon hirsutus            148.   0.457    -8.74
##  6 Silphium terebinthinaceum     171.   0.716    -4.53
##  7 Solidago juncea               130.   0.783    -7.79
##  8 Solidago nemoralis            122.   0.61     -7.84
##  9 Solidago rigida                86.5  1.1      -7.27
## 10 Solidago speciosa             263.   0.71     -8.27
## 11 Symphyotrichum laeve          127.   0.975    -7.76
## 12 Symphyotrichum oolentangiense 143.   0.362    -8.68
## 13 Tradescantia ohiensis         103.   0.921    -5.61
## 14 Zizia aptera                  100.   0.34     -6.42
```

### Finalize preparation

```
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year")

species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

## Statistics

### Descriptive analyses

```r
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 4 x 2
## # Groups:   year [4]
##    year     n
##   <dbl> <int>
## 1  2015    15
## 2  2016    18
## 3  2017    18
## 4  2018    18
```

## Ordinations

```r
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

## RLQ analyses

```r
m_rlq <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

### Global test

```r
randtest(m_rlq, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m_rlq, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test        Obs    Std.Obs   Alter Pvalue
## 1 Model 2 0.09506313 7.60634795 greater  0.001
## 2 Model 4 0.09506313 0.06333302 greater  0.428
```

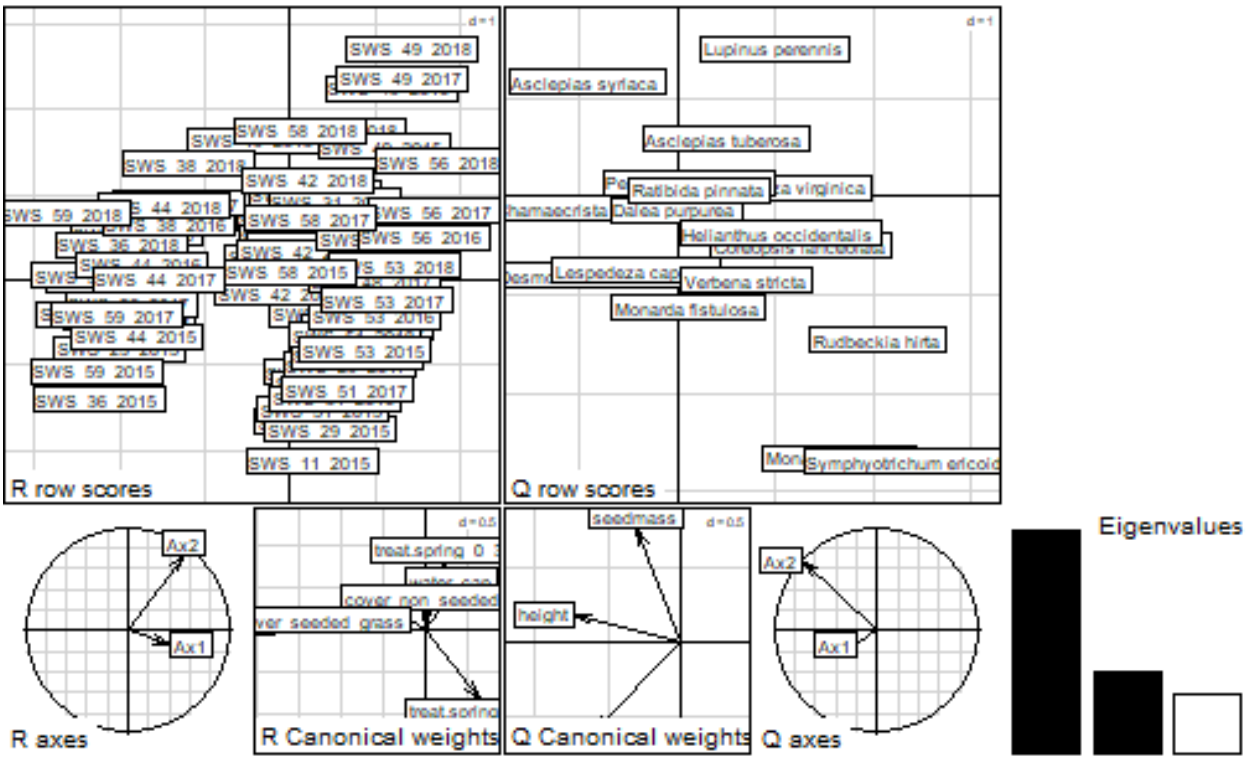
### Traits by environment crossed table

```
m_rlq$tab
```

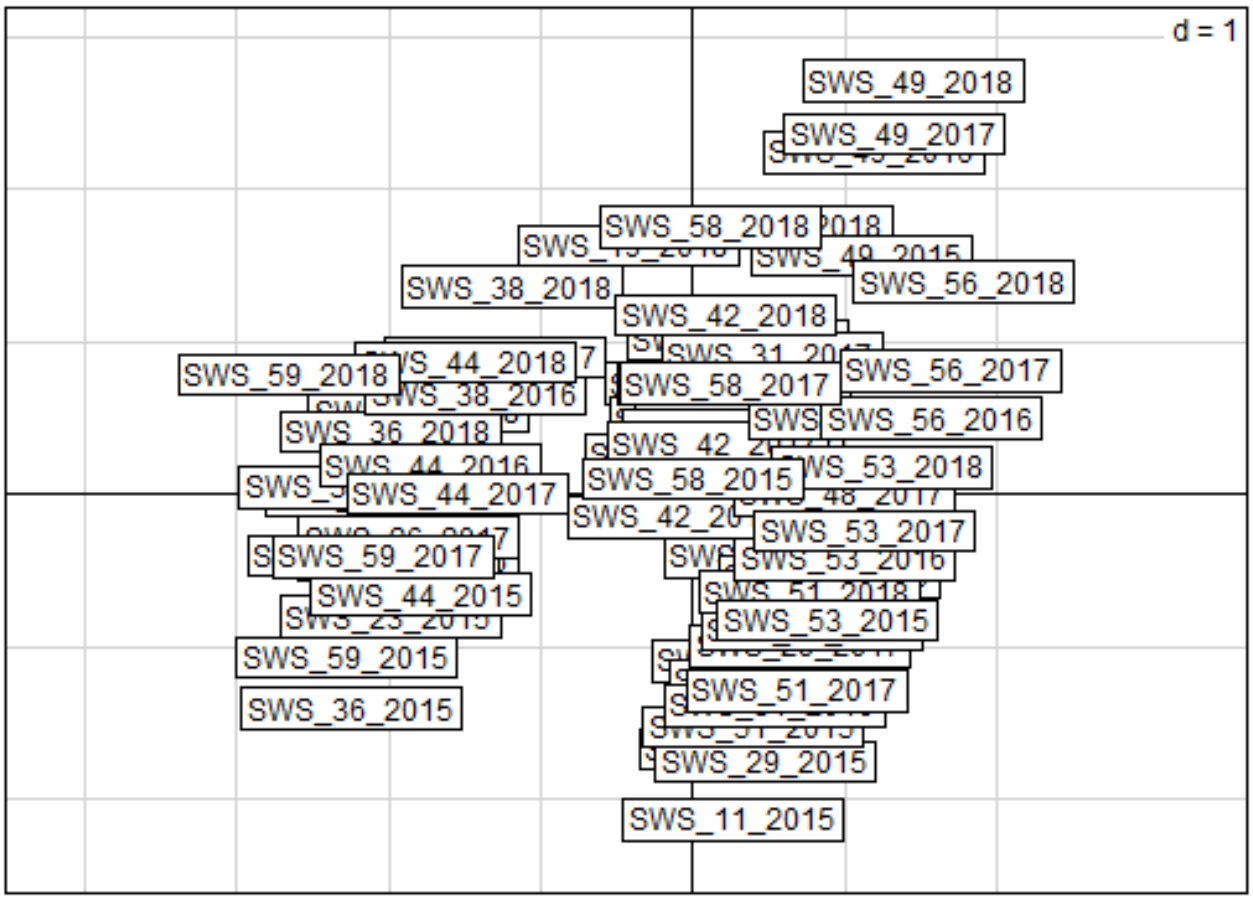
```
##                          sla      height     seedmass
## year               -0.01563389 -0.06214036  0.090760206
## treat.fall_0_33     0.24402927  0.32888521  0.128887563
## treat.spring_0_33  -0.10495269 -0.02667310  0.077447724
## treat.spring_1_33  -0.02350317 -0.14040976 -0.138299849
## water_cap          -0.11633551 -0.01086653 -0.006221456
## cover_non_seeded   -0.05264242  0.05761635 -0.010766242
## cover_seeded_grass  0.01671852  0.02924579  0.029693933
```

### RLQ Plots

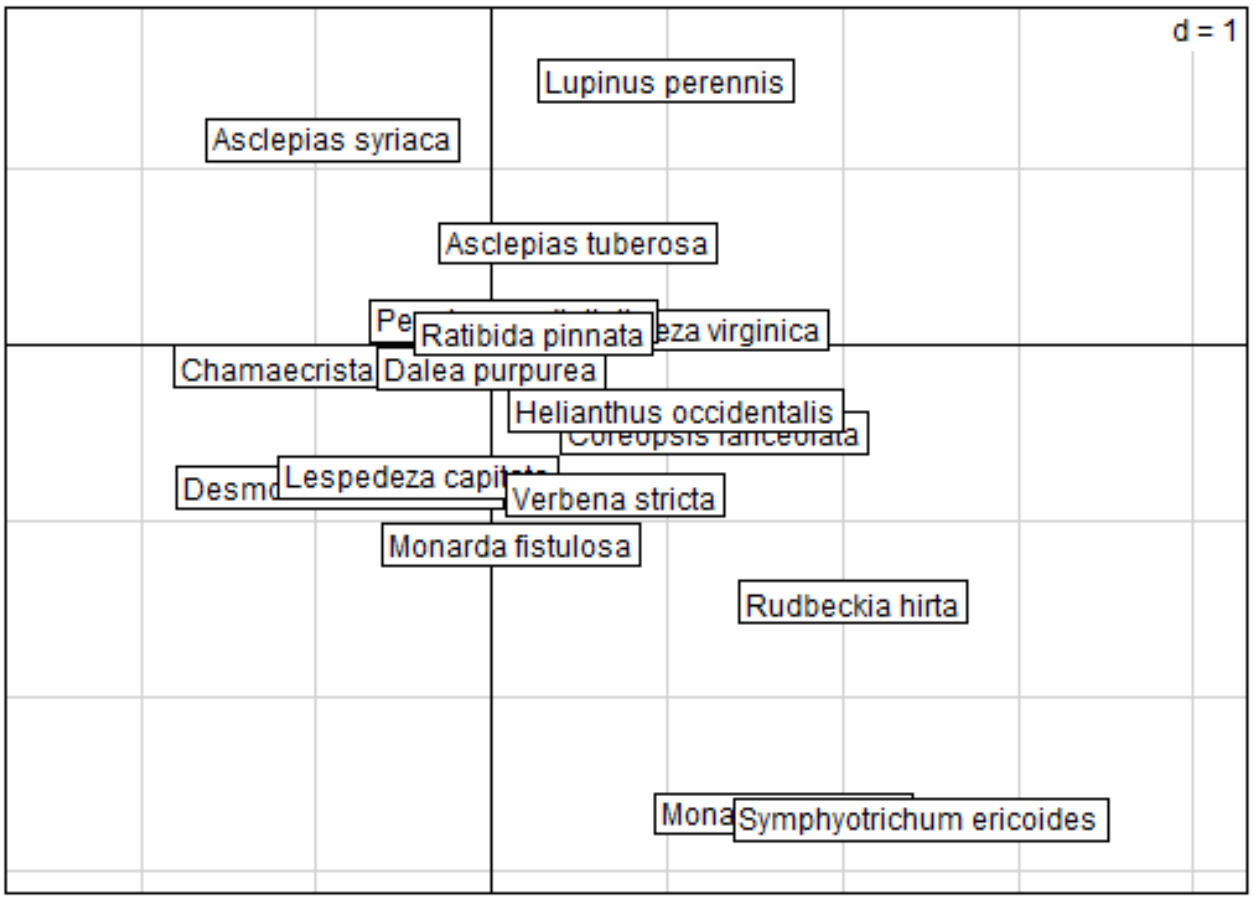
```
plot(m_rlq)
```



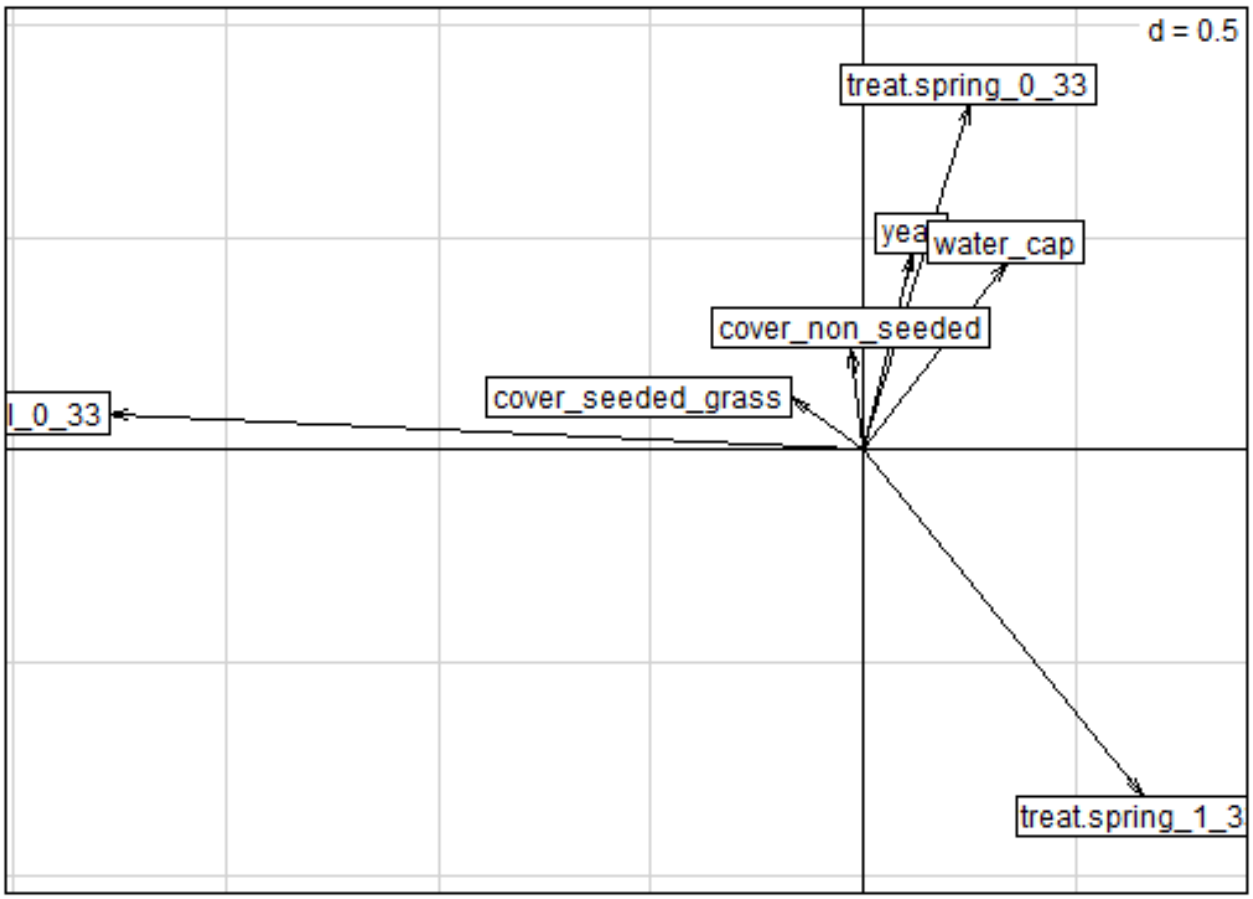

```
s.label(m_rlq$lR)
```



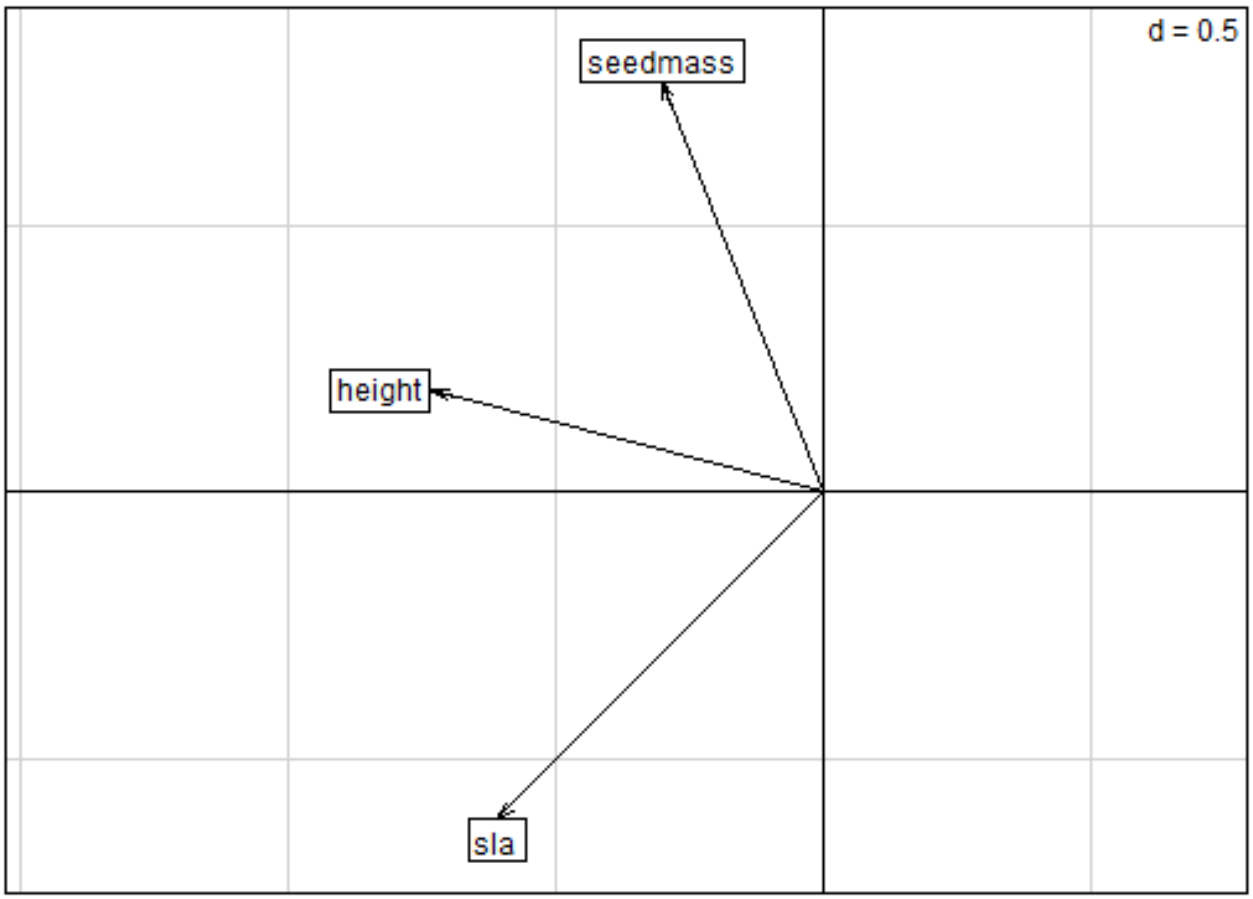

```
s.label(m_rlq$lQ)
```




```
s.arrow(m_rlq$l1)
```




```
s.arrow(m_rlq$c1)
```




### Fourth-corner analysis

```
m_4th <- fourthcorner(
  tabR = sites,
  tabL = species,
  tabQ = traits,
  modeltype = 6,
  p.adjust.method.G = "none",
  p.adjust.method.D = "none",
  nrepet = 49999
)
```

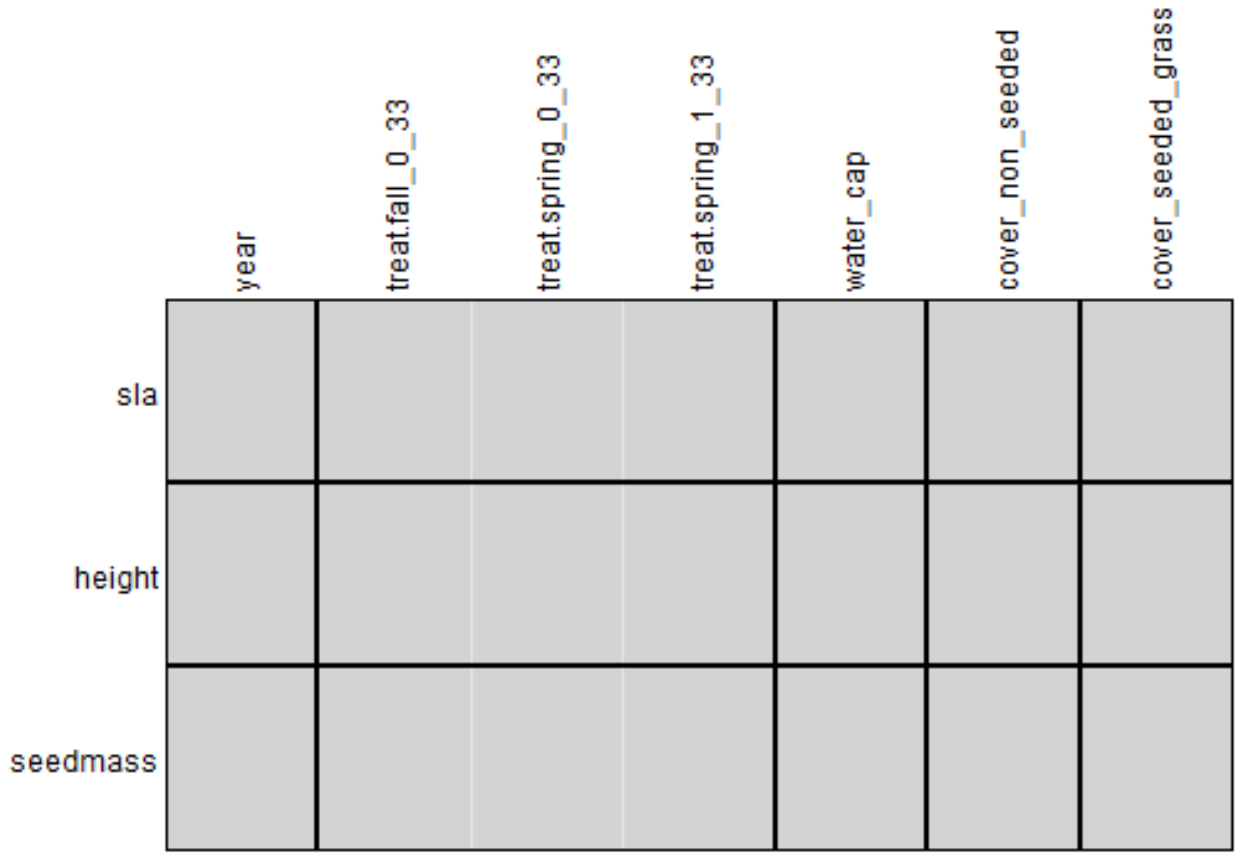
```
m_4th_adj <- p.adjust.4thcorner(
  m_4th,
  p.adjust.method.G = "fdr",
  p.adjust.method.D = "fdr",
  p.adjust.D = "global"
)
m_4th_adj
```

```
## Fourth-corner Statistics
## ------------------------
## Permutation method  Comb. 2 and 4  ( 49999  permutations)
##
## Adjustment method for multiple comparisons:   fdr
```

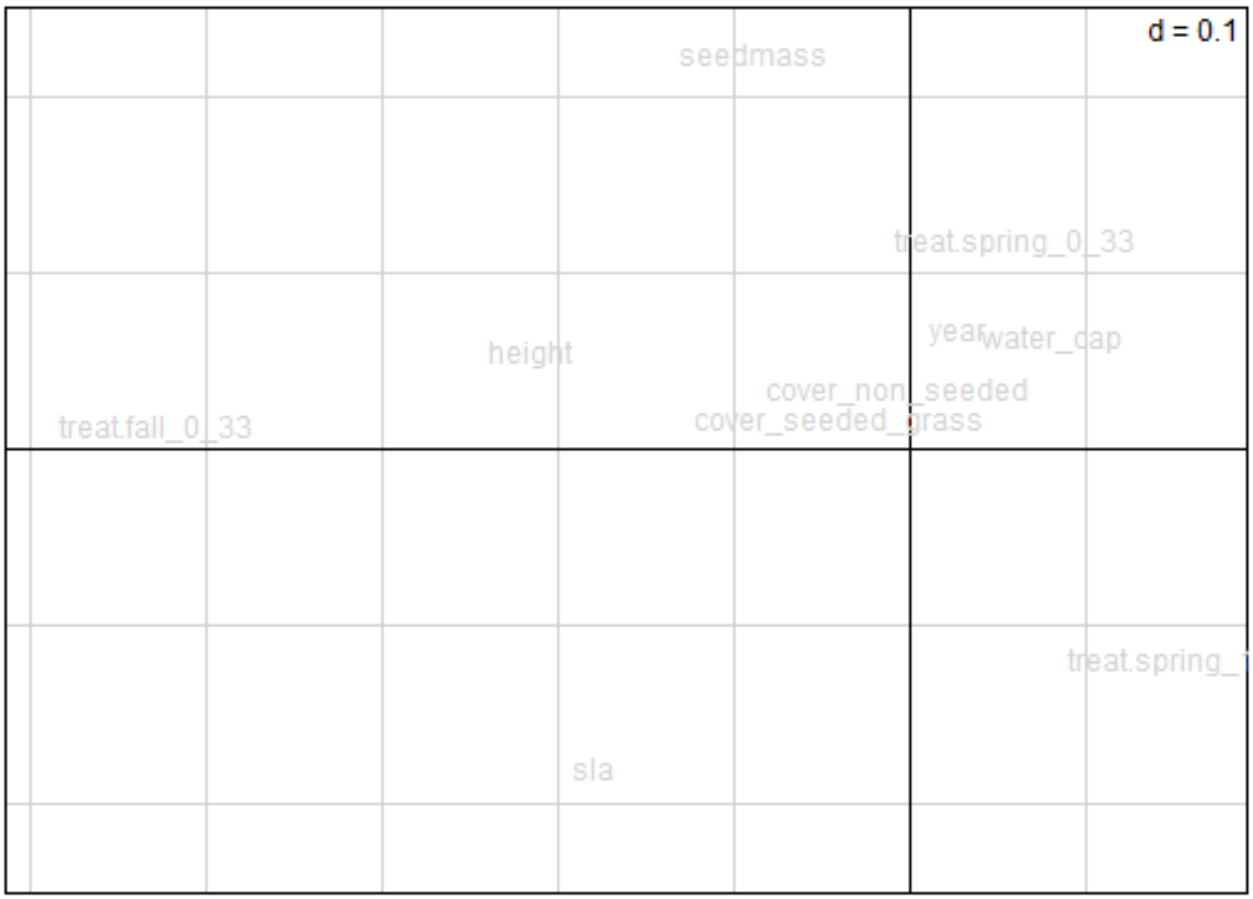
```
## call:  fourthcorner(tabR = sites, tabL = species, tabQ = traits, modeltype = 6,      nrepet = 49999,
##
## ---
##
##                              Test   Stat          Obs     Std.Obs     Alter
## 1                     year / sla      r -0.015633894 -0.26052879 two-sided
## 2          treat.fall_0_33 / sla Homog.  0.310237818  1.87817664      less
## 3        treat.spring_0_33 / sla Homog.  0.335790936 -0.40038186      less
## 4        treat.spring_1_33 / sla Homog.  0.337249661  0.16927417      less
## 5                water_cap / sla      r -0.116335507 -1.57924419 two-sided
## 6         cover_non_seeded / sla      r -0.052642418 -0.95433098 two-sided
## 7       cover_seeded_grass / sla      r  0.016718520  0.39426963 two-sided
## 8                  year / height      r -0.062140359 -1.17046706 two-sided
## 9       treat.fall_0_33 / height Homog.  0.232330559  0.97091206      less
## 10    treat.spring_0_33 / height Homog.  0.387597877  1.48942912      less
## 11    treat.spring_1_33 / height Homog.  0.349523061  0.63741617      less
## 12            water_cap / height      r -0.010866529 -0.15947423 two-sided
## 13     cover_non_seeded / height      r  0.057616352  1.07926953 two-sided
## 14   cover_seeded_grass / height      r  0.029245794  0.70468641 two-sided
## 15               year / seedmass      r  0.090760206  1.69802081 two-sided
## 16    treat.fall_0_33 / seedmass Homog.  0.212786165  0.53418477      less
## 17  treat.spring_0_33 / seedmass Homog.  0.403342735  1.14369329      less
## 18  treat.spring_1_33 / seedmass Homog.  0.370335388  0.77949657      less
## 19          water_cap / seedmass      r -0.006221456 -0.08530188 two-sided
## 20   cover_non_seeded / seedmass      r -0.010766242 -0.20232138 two-sided
## 21 cover_seeded_grass / seedmass      r  0.029693933  0.67873398 two-sided
##     Pvalue Pvalue.adj
## 1   0.8007    0.96098
## 2  0.96098    0.96098
## 3  0.34138    0.96098
## 4  0.56284    0.96098
## 5  0.11558    0.96098
## 6  0.36852    0.96098
## 7  0.69456    0.96098
## 8  0.25412    0.96098
## 9  0.82408    0.96098
## 10 0.93008    0.96098
## 11  0.7369    0.96098
## 12 0.88252    0.96098
## 13 0.30012    0.96098
## 14 0.51984    0.96098
## 15 0.08928    0.96098
## 16 0.71768    0.96098
## 17 0.87196    0.96098
## 18 0.77928    0.96098
## 19 0.93708    0.96098
## 20 0.84724    0.96098
## 21 0.50512    0.96098
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Plot**

```
plot(m_4th_adj, alpha = 0.05, stat = "D2")
```




```
plot(m_4th_adj,
  x.rlq = m_rlq,
  alpha = 0.05,
  stat = "D2",
  type = "biplot"
)
```




## Save

## Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.1
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.1     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.0.4    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.1         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.1        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```